Parallel beta-reduction is included in beta-reduction: if M bredpar M' then M bred M'

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: all four rule cases are expanded in the span, each producing its reduction sequence explicitly; there is no omitted or delegated step.

Editorial note. The label lem:par-red names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Editorial note. OLP reuses this label inside the unit; the card is identified by its statement span (source lines 496-498) and the claim key carries the statement start line.

Allowed standard prerequisites (editorial).

- Inductive definition of parallel beta-reduction. Same unit, rules \ollabel{defn:bredpar1}-\ollabel{defn:bredpar4} (section parallel-beta-reduction, definition at source line 773). The four numbered cases of the derivation induction, each converted into an explicit reduction sequence.

Source reference. Original definition defn-bredpar1: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 770–770. Prerequisite pointer invoked at proof line(s) 502; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition defn-bredpar2: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 771–771. Prerequisite pointer invoked at proof line(s) 504; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition defn-bredpar3: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 773–773. Prerequisite pointer invoked at proof line(s) 509; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition defn-bredpar4: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 775–775. Prerequisite pointer invoked at proof line(s) 514; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *If $M \to_\beta M'$, then $M \to_\beta M'$.*

Proof. By induction on the !!derivation of $M \to_\beta M'$.

1. The last rule is defn-bredpar1: Then M and M' are just x, and $x \to_\beta x$.

2. The last rule is defn-bredpar2: M is $\lambda[x][N]$ and M' is $\lambda[x][N']$ for some x, N, N', where $N \to_\beta N'$. By induction hypothesis we have $N \to_\beta N'$. Then $\lambda[x][N] \to_\beta \lambda[x][N']$ (by the same series of $=_\beta$ contractions as $N \to_\beta N'$).

3. The last rule is defn-bredpar3: M is PQ and M' is $P'Q'$ for some P, Q, P', Q', where $P \to_\beta P'$ and $Q \to_\beta Q'$. By induction hypothesis we have $P \to_\beta P'$ and $Q \to_\beta Q'$. So $PQ \to_\beta P'Q'$ by the reduction sequence $P \to_\beta P'$ followed by the reduction $Q \to_\beta Q'$.

4. The last rule is defn-bredpar4: M is $(\lambda[x][N])Q$ and M' is $N'[Q'/x]$ for some x, N, M', Q, Q', where $N \to_\beta N'$ and $Q \to_\beta Q'$. By induction hypothesis we get $Q \to_\beta Q'$ and $N \to_\beta N'$. So $(\lambda[x][N])Q \to_\beta N'[Q'/x]$ by $N \to_\beta N'$ followed by $Q \to_\beta Q'$ and finally contraction of $(\lambda[x][N'])Q'$ to $N'[Q'/x]$.

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